

HOMS

Learner Worksheet

Motion in One Dimension

Physical Sciences | Grade 10 | Term 3

Complete every question. Total: 40 marks.

Learner name	
Date	

CAPS-aligned draft. Educator or subject-expert approval is required before classroom use.

Instructions: Show formulae, substitutions, units and direction where required. Use the supplied graphs for graph questions.

Question 1: Concepts

1.1 Define a frame of reference. (2)

1.2 Distinguish between distance and displacement. (2)

1.3 Explain why a negative velocity does not necessarily mean that an object is slowing down. (2)

1.4 State the physical meaning of the gradient of a position-time graph. (1)

1.5 State the physical meaning of the area under a velocity-time graph. (1)

Question 2: Calculations

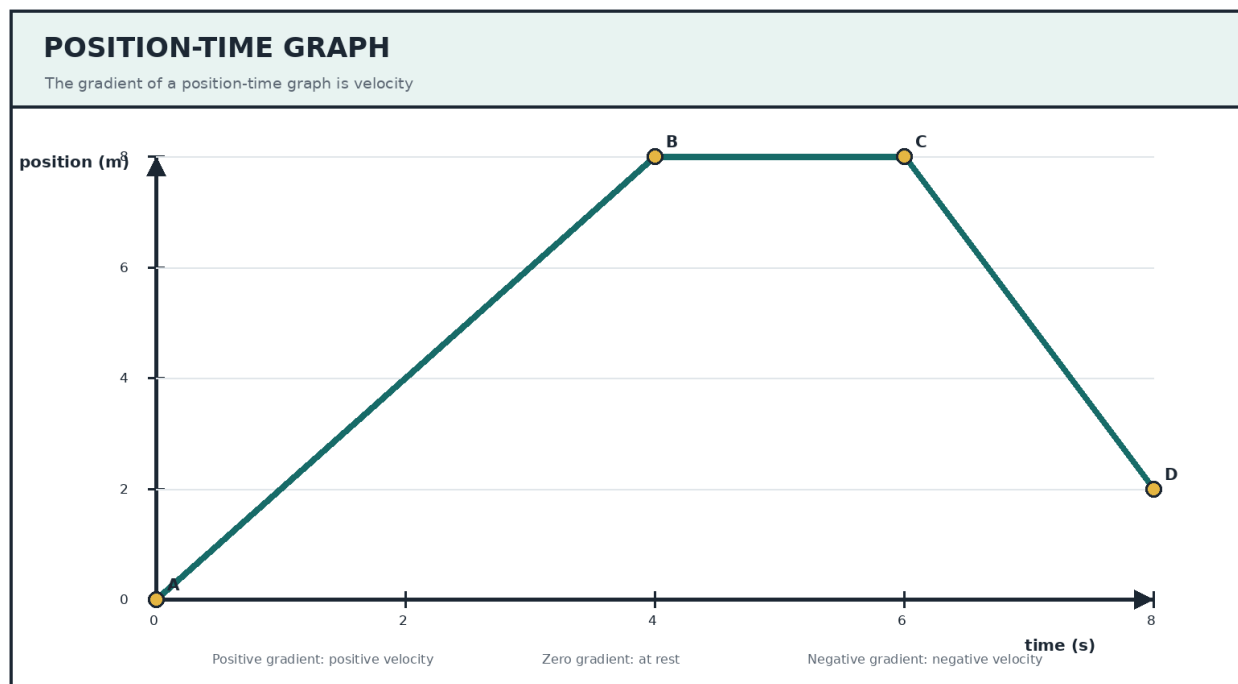
2.1 A learner walks 60 m east and then 25 m west. Calculate the distance and displacement. (4)

2.2 The journey in Question 2.1 takes 50 s. Calculate average speed and average velocity. (4)

2.3 A car changes velocity from $+4 \text{ m.s}^{-1}$ to $+16 \text{ m.s}^{-1}$ in 6 s. Calculate its acceleration. (3)

2.4 A bicycle moving at 10 m.s^{-1} slows uniformly at 2 m.s^{-2} for 3 s. Calculate its final velocity. (3)

Question 3: Graph Interpretation



3.1 Refer to the position-time graph. Calculate the velocity from A to B.

(3)

3.2 Describe the motion from B to C.

(2)

3.3 Calculate the velocity from C to D.

(3)

3.4 Explain what the negative answer in Question 3.3 means.

(2)

Question 4: Uniform Acceleration

4.1 A trolley starts at 3 m.s^{-1} and accelerates uniformly at 2 m.s^{-2} for 5 s. Calculate its final velocity.

(3)

4.2 Calculate the displacement of the trolley during the 5 s interval.

(5)

TOTAL: 40